

Paul Hondola

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EDUCATION

Bachelor of Science in Computer Engineering

Oct. 2023 – Jun. 2027

Universitatea Politehnica Timisoara

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Computer Architecture, Distributed Systems, Networking, Web Development.

EXPERIENCE

Software Engineer Intern

Mar. 2026 – Present

Nokia | Python, Jenkins, Docker, Power BI

- Profiled and optimized data-processing Python scripts, reducing end-to-end runtime by 50% through algorithmic refactoring and numpy vectorization.
- Contributed to the Mobile Infrastructure Metrics Automation platform, developing ETL scripts and automated test pipelines for various analytics.
- Enhanced CI/CD pipelines using Jenkins and Docker for scalable metric processing and analysis.
- Maintained and improved interactive Power BI dashboards to surface key metrics and KPIs.
- Contributed reusable modules to the team's shared core library, extending common functionality leveraged across multiple internal automation projects.

Full Stack Developer Intern

Jul. 2025 – Oct. 2025

Hibyte | Typescript, Angular, NestJS, Payload CMS, Supabase

- Engineered 'GameBox', a full-stack monorepo for game center management, integrating a customer-facing web app with an admin CMS.
- Built a responsive frontend architecture using Angular, TypeScript, and SCSS.
- Developed a backend system using NestJS and Payload CMS, leveraging Supabase for secure authentication and real-time database operations.

Malware Analyst Trainee

Apr. 2025 – Jun. 2025

Bitdefender | Java, Python, C, x86 Assembly, IDA Pro, jadx

- Conducted static and dynamic analysis of Windows and Android malware samples to identify command-and-control protocols and persistence mechanisms.
- Reverse engineered ransomware encryption routines using IDA Pro (x86 disassembly) and jadx (Java decompilation) to understand payload behavior.
- Investigated Android system vulnerabilities and exploitation techniques, analyzing APK structures and security models in sandboxed environments.

TECHNICAL SKILLS

Languages: C/C++, C#, Python, Java, TypeScript, SQL, Bash

High Performance Computing: MPI, OpenMP, POSIX Threads, SIMD, Shared Memory, Distributed Systems

Systems and Tools: Linux, Docker, Jenkins, Git, Meson, CMake, Postman, IDA Pro, jadx, Wireshark

Web Development: React, Angular, NestJS, ASP.NET Core, Hono

Databases: PostgreSQL, Supabase, Entity Framework Core

Data Analysis: NumPy, Pandas, Matplotlib, PyTorch, Power BI

Spoken Languages: English (C1/Advanced), Romanian (Native)

PROJECTS

Vibecodium - Collaborative Browser-Based IDE | *React, Hono, Bun, Supabase, Docker, Vercel* Mar. 2026 – Present

- Won 1st Place in Web Development at the iTEC 2026 Hackathon.
- Built a collaborative browser-based IDE with real-time multi-user editing (Yjs CRDT).
- Integrated an AI coding assistant using Server-Sent Events (SSE) for real-time code generation and completion suggestions.
- Implemented a sandboxed Docker execution environment for running user-submitted code with isolated filesystem and network access.

- Integrated one-click Vercel deployment for user projects with automatic environment variable configuration and build pipeline orchestration.

Smart Shopping Assistant | *React, TypeScript, ASP.NET Core, PostgreSQL, OpenAI* Apr. 2026 – Jun. 2026

- Built a full-stack e-commerce platform with an AI-powered agentic cart analyzer, developed within the LigaAC Labs x NetRom industry mentorship program.
- Architected a sequential two-agent LLM pipeline that detects active and near-miss promotions and composes personalized product recommendations.
- Engineered a layered ASP.NET Core backend over PostgreSQL with Entity Framework Core and JWT role-based authentication.
- Kept a deterministic server-side promotion engine authoritative over the LLM to ensure pricing accuracy.
- Developed a responsive React 19 + TypeScript SPA (Vite, Material-UI, TanStack Query) consuming the REST API.

ArchLens Analyzer | *Python, Java, C#, Reverse Engineering* Mar. 2026 – Apr. 2026

- Polyglot CLI tool (Python orchestrator + Java/C# backends) that reverse-engineers compiled binaries (.jar, .dll) into UML class diagrams using runtime reflection.
- Outputs PlantUML and yUML; built with a layered pipeline architecture across a Maven and .NET monorepo.

AtomC Compiler | *C, Meson, Compiler Design* Mar. 2026 – May. 2026

- Developed a C-like language compiler frontend from scratch in C.
- Implemented a lexer, parser, and semantic analyzer to process and validate C source code.
- Established CI pipeline with integration test suites for lexer, parser, domain/type analyzer, stack-based virtual machine and code generator.

Distributed Image Processing System | *C, MPI, OpenMP, Meson, Python, Assembly* Dec. 2025 – Jan. 2026

- Implemented a high-performance hybrid parallel image processing engine utilizing MPI for distributed memory and OpenMP for shared memory parallelism.
- Engineered diverse parallel architectures including a standard multithreaded approach, distributed message passing, shared filesystem optimization, a producer-consumer task pool model and hybrid parallelism.
- Validated the efficiency of SIMD vectorization by inspecting generated assembly code across different compiler optimization levels.
- Optimized computationally intensive convolution kernels (Gaussian Blur, Ridge Detection) for processing large BMP images across cluster nodes.
- Developed an automated benchmarking suite with Python visualization to analyze strong scaling, speedup, and parallel efficiency.

Parallel DMM Benchmark and Analysis | *C, OpenMP, Meson, Python, Assembly* Oct. 2025 – Nov. 2025

- Developed a benchmarking suite for dense matrix multiplication, implementing serial and parallel algorithms with OpenMP.
- Implemented loop permutations and cache blocking (tiling) strategies to optimize memory access patterns and minimize cache misses.

Deepblue Maker - Underwater Robotics Summer Camp Jul. 2025 – Aug. 2025

Hangzhou Dianzi University, China

- Co-engineered a remotely operated underwater vehicle (ROV) capable of environmental mapping and real-time object detection.
- Mechanical: Designed and 3D-printed a custom water-sealed chassis using SolidWorks; validated buoyancy and structural integrity through physical testing.
- Embedded Systems: Programmed an STM32 microcontroller for precise thruster propulsion control and hardware interfacing.
- Computer Vision: Deployed a YOLO-based object detection model on a Raspberry Pi 5 to identify sea life from live camera feeds.